# Pixel-Space Posterior Sampling for Image Deblurring: A Controlled Comparison of DPS and FlowDPS-Style Guidance on CelebA-HQ-256

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## Abstract

Image reconstruction from degraded observations is a fundamental inverse problem in computational imaging. *Diffusion Posterior Sampling* (DPS) Chung et al. (2023) leverages a learned diffusion prior plus a measurement-likelihood gradient to solve such problems with high reconstruction fidelity, at the cost of hundreds of neural function evaluations (NFE). *Flow Matching* models Lipman et al. (2023); Liu et al. (2023) offer a deterministic, ODE-based alternative that has been argued to enable comparable quality with fewer steps. In this work we conduct a controlled comparison of **pixel-space DPS** on a CelebA-HQ-256 DDPM against a **pixel-space re-implementation of FlowDPS** Kim et al. (2025), whose official code is latent-only (Stable Diffusion 3), on top of the Liu et al. 2023 Rectified Flow CelebA-HQ-256 checkpoint. We evaluate both methods on Gaussian deblurring across $3 \times 2 \times 3 = 18$ conditions $\times$ $50$ images $= 1800$ reconstructions, and study operator-mismatch robustness using motion-blur measurements while the sampler's likelihood assumes Gaussian (1200 additional reconstructions). We further include Wiener-filter and DnCNN+PnP-ADMM baselines (600 reconstructions). On the matched-Gaussian grid Pixel-DPS dominates the comparison, with a PSNR advantage of $+1.4$ to $+4.2$ dB over our FlowDPS reimplementation, and FlowDPS saturates with NFE while Pixel-DPS keeps improving. Under operator mismatch, however, FlowDPS-on-RF is *relatively more robust* (degrades by $\sim$2.5 dB versus $\sim$5 dB for Pixel-DPS), reversing the matched-condition comparison. All four methods comfortably beat Wiener; DnCNN+PnP-ADMM beats our FlowDPS reimplementation in $5/6$ noisy cells while running $\sim$60$\times$ faster than DPS. We additionally evaluate *thirteen* publication-improvement candidates spanning EMA-aligned weight loading, time-varying $\zeta$ schedules, spectral-aware likelihood reweighting, $\Pi$-GDM-style covariance correction (heuristic and analytical), a 2nd-order Heun integrator at matched and unconstrained NFE budgets, and two gradient-free particle samplers (constrained-particle and tempered SMC). **Three candidates** clear a strict Bonferroni-corrected significance gate: `flowdps_rf_v2` adds $+0.77$ dB (EMA-aligned weight load + ramp-$\zeta$, $6/6$ cells significant); `pixel_dps_spectral` adds $+0.46$ dB ($5/6$ cells) on the matched-Gaussian grid; and `flowdps_rf_spectral` adds $+0.54$ dB ($5/6$ cells at $n$=100). The two spectral wins show a *paradoxical reversal* relative to the Tier-B robustness study: the same spectral-weight algorithm *fails* under operator mismatch and *wins* on matched-operator data, isolating the mechanism as "downweight residuals in noise-dominated frequency bands of the *assumed* operator". The remaining ten variants are documented as instructive negative results or near-misses with diagnoses, including a DPS-vs-$\Pi$-GDM L2-norm/squared convention mismatch, a guidance-saturation diagnosis from a Heun integrator at $2\times$ NFE, and a path-degeneracy diagnosis from gradient-free SMC.

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The findings give concrete guidance for choosing between these methods under known versus uncertain forward operators.

# 1 Introduction

Image reconstruction from degraded observations is a fundamental problem in computer vision and computational imaging. It is typically formulated as an inverse problem,

$$y = Ax + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \sigma_n^2 I), \tag{1}$$

where $A$ is a known or partially known degradation operator (e.g. Gaussian blur), and the goal is to recover the unknown clean image $x$ given only $y$ and a prior over natural images. The problem is inherently ill-posed; multiple plausible $x$ can produce the same observation, so a strong prior is essential. Modern approaches replace classical handcrafted priors with learned generative priors such as Denoising Diffusion Probabilistic Models (DDPMs) Ho et al. (2020) and Flow Matching models Lipman et al. (2023); Liu et al. (2023). These priors combined with explicit measurement guidance form the basis of methods like DPS Chung et al. (2023), DDRM Kawar et al. (2022), and the more recent FlowDPS Kim et al. (2025).

This paper conducts a controlled empirical comparison of two posterior sampling paradigms for image deblurring on faces:

1. **Pixel-DPS**: the original DPS sampler from Chung et al. Chung et al. (2023), on a CelebA-HQ-256 DDPM checkpoint (`google/ddpm-ema-celebahq-256`).
2. **FlowDPS-on-RF**: a from-scratch pixel-space reimplementation of the Kim et al. 2025 FlowDPS algorithm on top of the Liu et al. 2023 CelebA-HQ-256 Rectified Flow checkpoint Liu et al. (2023). The official Kim 2025 codebase is Stable Diffusion 3-only; our reimplementation exists so we can do a true pixel-space comparison against Pixel-DPS at the same resolution and on the same dataset.

Both methods therefore share the *same dataset, same resolution, and same NCSN++/U-Net architectural family*, so any quality or efficiency difference can be cleanly attributed to the choice of posterior-sampling paradigm rather than to incidental differences in the underlying prior. We additionally evaluate two classical baselines (Wiener, DnCNN+PnP-ADMM) and one operator-mismatch robustness study (true measurement is motion blur; sampler's likelihood assumes Gaussian).

The study is guided by four research questions:

1. **Quality.** Can a flow-based posterior sampler (FlowDPS on Rectified Flow) match diffusion-based DPS in reconstruction fidelity (PSNR / SSIM / LPIPS) at matched NFE budgets?
2. **Efficiency.** How does per-image wall-clock scale with NFE for each method, and is there a regime where the deterministic flow trajectory yields a measurable speed advantage?
3. **Robustness.** Under forward-model misspecification (true measurement is motion blur, sampler's likelihood assumes Gaussian), which method degrades more gracefully?
4. **Baselines.** How do the diffusion / flow posterior samplers compare to classical (Wiener) and supervised (DnCNN+PnP-ADMM) reconstruction baselines on identical inputs?

The contributions of this work are:

- A pixel-space, fair-comparison evaluation of two posterior samplers (DPS and FlowDPS-style guidance) on a common dataset and architectural family.
- Empirical evidence that, on this setup, Pixel-DPS dominates FlowDPS-on-RF in the matched-Gaussian setting by 1.4–4.2 dB PSNR, while FlowDPS-on-RF degrades less under operator mismatch — reversing the matched-condition ordering.
- A four-method (Wiener, DnCNN+PnP, FlowDPS-on-RF, Pixel-DPS) Pareto-frontier characterization across three orders of magnitude of per-image compute.
- A rigorous *ablation study* of thirteen candidate improvements to either sampler — EMA-aligned weight loading, time-varying $\zeta$ schedules (fixed and noise-adaptive), spectral-domain residual

reweighting (on robustness and matched grids), Π-GDM-style covariance correction (heuristic and analytical), a 2nd-order Heun integrator at both matched and unconstrained NFE budgets, and gradient-free particle samplers (constrained-particle and tempered SMC) — evaluated against a strict Bonferroni-corrected significance gate. We report *three confirmed wins*: `flowdps_rf_v2` (EMA + ramp-$\zeta$, $+0.77\,$dB, $6/6$ cells), `pixel_dps_spectral` ($+0.46\,$dB, $5/6$ cells matched), and `flowdps_rf_spectral` ($+0.54\,$dB, $5/6$ cells matched at $n$=100). The Tier-B spectral mechanism shows a *paradoxical reversal*: the same algorithm fails on the operator-mismatch grid and wins on the matched grid, sharpening its mechanistic interpretation. Ten candidates are documented as near-misses or diagnosed failures, including a DPS L2-norm vs. Π-GDM L2-squared convention mismatch, a guidance-saturation diagnosis from Heun at $2\times$ NFE, and a path-degeneracy diagnosis from gradient-free SMC.

- A reproducible open-source pipeline at `https://github.com/adnan821/dps_flowdps`, including the 3,600 measured rows, all per-experiment logs, and the full LaTeX source for this report.

## 2 Related Work

Classical inverse-problem solvers rely on handcrafted regularizers and analytical priors. The Wiener filter solves the linear MMSE problem in the Fourier domain for Gaussian degradations and remains a fast, interpretable baseline. Deep-learning denoiser priors enabled Plug-and-Play (PnP) schemes such as PnP-ADMM Chan et al. (2017); Zhang et al. (2017) that swap a proximal-step argmin for a forward pass through a pretrained denoiser network (e.g. DnCNN, DRUNet). Diffusion-based posterior sampling generalized this idea by injecting a measurement-consistency term into the reverse diffusion process: DDRM Kawar et al. (2022) uses spectral projection in the SVD basis of $A$, and DPS Chung et al. (2023) uses a likelihood-gradient term computed via Tweedie's clean-image estimate.

Flow Matching and Rectified Flow Lipman et al. (2023); Liu et al. (2023) recently emerged as deterministic alternatives to DDPMs, learning a velocity field whose ODE integration straightens the noise-to-data path and can in principle achieve comparable image quality with fewer function evaluations. FlowDPS Kim et al. (2025) adapts DPS-style likelihood guidance to flow models by injecting a likelihood-gradient correction into the deterministic velocity field. The official FlowDPS implementation is built on the SD3 latent transformer flow model Esser et al. (2024) and operates at $768 \times 768$ in latent space.

Beyond DPS-style likelihood-gradient guidance, several recent works explore alternative posterior-sampling regimes for diffusion- and flow-based priors. Meanti et al. Meanti et al. (2025) cast the inverse problem as distribution matching between a parameterized sampler and a target conditioned on the measurement, avoiding explicit likelihood gradients. Dou et al. Dou et al. (2026) propose a *gradient-free* constrained-particle approach that solves diffusion inverse problems using only forward passes through the score / flow network, side-stepping the cost of backpropagating through the model that DPS-style methods incur. Dasgupta et al. Dasgupta et al. (2026) study *physics-constrained* conditional flow matching for inverse problems with hard physical constraints (e.g. PDE residuals), where posterior-sample physical validity matters as much as image quality. Our work uses gradient-based DPS / FlowDPS guidance; we leave a gradient-free or physics-constrained variant to future work.

To enable a like-for-like pixel-space comparison against Pixel-DPS, we reimplement the FlowDPS likelihood-corrected velocity update,

$$v_{\text{eff}}(x_t, t) = v_\theta(x_t, t) - \zeta \nabla_{x_t} \|y - A(\hat{z}_1(x_t, t))\|_2, \tag{2}$$

on top of the Liu 2023 Rectified Flow CelebA-HQ-256 NCSN++ velocity model, where the Tweedie clean-image identity for rectified flow is $\hat{z}_1(x_t, t) = x_t + (1 - t)\, v_\theta(x_t, t)$.

## 3 Methodology

**Forward model.** $y = Ax + \epsilon$, with $A$ either a Gaussian blur of standard deviation $\sigma_b \in \{1.5, 3.0, 5.0\}$ (main grid) or a motion blur motion$(L, \theta)$ with $L \in \{15, 25, 35\}$ pixels and $\theta \in \{0^\circ, 45^\circ\}$ (robustness study), and $\epsilon \sim \mathcal{N}(0, \sigma_n^2 I)$ with $\sigma_n \in \{0.0, 0.05\}$.

**Pixel-DPS.** We adopt the DDPM reverse process with DDIM sampling Song et al. (2021). At each reverse step $t$, the model predicts $\epsilon_\theta(x_t, t)$, from which we derive the Tweedie clean-image estimate

$$\hat{x}_0(x_t) = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\,\epsilon_\theta(x_t,t)}{\sqrt{\bar{\alpha}_t}}. \tag{3}$$

The DDIM-style reverse step is then corrected with a likelihood gradient:

$$x_{t-1} = x_{t-1}^{\text{DDIM}} - \zeta \nabla_{x_t} \|y - A(\hat{x}_0(x_t))\|_2. \tag{4}$$

Two implementation choices matter substantially. First, we use the **L2 norm** (not the squared L2) of the residual; squaring the residual and dividing by $\sigma_n^2$ blows up the gradient and prevents convergence (verified empirically in our EXP-001 ablation). Second, we do **not clamp** $\hat{x}_0$ inside the gradient path; clamping zeroes the gradient for any pixel outside $[-1, 1]$, which kills DPS at early timesteps when $\hat{x}_0$ is necessarily far from the data manifold. Clamping is applied only at the final returned tensor.

**FlowDPS-on-RF.** We use the Liu 2023 CelebA-HQ-256 1-Rectified Flow checkpoint, which provides a learned velocity field $v_\theta(x_t, t)$ over continuous time $t \in [0, 1]$, with $x_0 \sim \mathcal{N}(0, I)$ at $t = 0$ and $x \sim p_{\text{data}}$ at $t = 1$. Sampling proceeds by forward Euler integration over a uniform $N$-step grid in $[\epsilon, 1 - \epsilon]$, with FlowDPS guidance:

$$v_{\text{eff}}(x_t, t) = v_\theta(x_t, t) - \zeta \nabla_{x_t} \|y - A(\hat{z}_1(x_t, t))\|_2, \qquad x_{t+dt} = x_t + dt \cdot v_{\text{eff}}(x_t, t). \tag{5}$$

The clean-image estimate $\hat{z}_1 = x_t + (1 - t)v_\theta(x_t, t)$ is RF's Tweedie identity. Our reimplementation lives in `src/samplers/flowdps_rf.py` in the project repository.

**Baselines.**

- **Wiener filter:** per-channel 2-D Wiener deconvolution in the Fourier domain, with noise-to-signal scalar $K = 10^{-3}$ for $\sigma_n = 0$ and $K = 5 \cdot 10^{-2}$ for $\sigma_n = 0.05$ (tuned on a 2-image validation).
- **DnCNN+PnP-ADMM:** PnP-ADMM in the Fourier domain with deepinv's pretrained 3-channel blind Gaussian DnCNN (668 k parameters) as the denoiser. 24 outer iterations, $\rho = 2.0$, descending denoiser-$\sigma$ schedule with $\sigma_{\text{max}} = \max(0.10, 4\sigma_n + 0.05)$ and $\sigma_{\text{min}} = \max(0.02, \sigma_n)$.

**Implementation details.** All experiments run on a single **NVIDIA RTX 3060 12 GB** GPU. Models load via Hugging Face's `diffusers` 0.30.1 on PyTorch 2.4.1+cu121. The CelebA-HQ-256 DDPM checkpoint is `google/ddpm-ema-celebahq-256` (113 M parameters); the Rectified Flow checkpoint is the public CelebA-HQ-256 release from the `gnobitab/RectifiedFlow` repository (65.5 M parameters, NCSN++ architecture, JIT-compiled `upfirdn2d` CUDA op). LPIPS uses the AlexNet backbone via the standard `lpips` package. Wall-clock timings use CUDA events.

## 4 Experimental Design

**Dataset.** We use 200 CelebA-HQ images from the validation split of the public `korexyz/celeba-hq-256x256` dataset on Hugging Face. The first 50 images are used for the main experimental grid and the robustness study. The 200-image set size is set by our compute envelope on the RTX 3060 12 GB; we disclose this explicitly so per-cell error bars reflect this sample size.

**Hyperparameter tuning.** Guidance scale $\zeta$ tuned on a 2-image held-out subset (EXP-001 for Pixel-DPS, EXP-006/007 for FlowDPS-on-RF). Locked values for the **v1** baselines: $\zeta = 10$ for Pixel-DPS, $\zeta = 100$ for FlowDPS-on-RF. Both are monotonic-best in their respective ranges; values past these begin to hurt quality. The eight publication-improvement variants in §5.7 each carry their own re-tuned $\zeta$ (or $\zeta$-schedule) chosen on a disjoint held-out subset (images 50–54), to avoid leakage into the 50-image evaluation set used for the significance tests.

**Metrics.** We report PSNR (dB, on the $[0, 1]$ range), SSIM (a PyTorch-native implementation matching the conventional Gaussian-windowed formula), and LPIPS (AlexNet backbone via the standard `lpips` package). Higher is better for PSNR / SSIM; lower for LPIPS. Computational efficiency is reported as wall-clock seconds per image (CUDA-event timed, excluding I/O) and the number of neural function evaluations (NFE) per reconstruction. We omit Kernel Inception Distance (KID), which the proposal listed: distributional metrics over 50 samples per cell are too noisy to be informative, and our paired-image PSNR / SSIM / LPIPS already capture the same quality axes.

Table 1: Experimental configuration matrix.

| Parameter | Values | Cells |
|---|---|---|
| Gaussian blur $\sigma_b$ | $\{1.5, 3.0, 5.0\}$ | 3 |
| Additive noise $\sigma_n$ | $\{0.0, 0.05\}$ | 2 |
| NFE budget | $\{25, 50, 100\}$ | 3 |
| Method | {Pixel-DPS, FlowDPS-on-RF} | 2 |
| Total main-grid conditions | $3 \cdot 2 \cdot 3 \cdot 2 = 36$ | — |
| Images per cell | 50 | — |
| Total reconstructions (main grid) | 1,800 | — |
| Robustness: motion length $L$ | $\{15, 25, 35\}$ | 3 |
| Robustness: motion angle $\theta$ | $\{0°, 45°\}$ | 2 |
| Robustness: NFE | 50 (fixed) | — |
| Total robustness reconstructions | $3 \cdot 2 \cdot 2 \cdot 2 \cdot 50 = 1{,}200$ | — |

**Baselines.** To contextualize the DPS / FlowDPS comparison we include two non- diffusion baselines:

- **Wiener filter**: per-channel 2-D Wiener deconvolution in the Fourier domain, with noise-to-signal scalar $K = 10^{-3}$ for $\sigma_n = 0$ and $K = 5 \cdot 10^{-2}$ for $\sigma_n = 0.05$. Evaluated on the 6-cell Gaussian grid (300 reconstructions, $\sim$16 ms per image on CPU).
- **DnCNN+PnP-ADMM**: Plug-and-Play ADMM in the Fourier domain with deepinv's pretrained 3-channel blind Gaussian DnCNN (668 k parameters) Zhang et al. (2017); Chan et al. (2017). 24 outer iterations, $\rho = 2.0$, descending denoiser-$\sigma$ schedule with $\sigma_{\text{max}} = \max(0.10, 4\sigma_n + 0.05)$ and $\sigma_{\text{min}} = \max(0.02, \sigma_n)$. 300 reconstructions, $\sim$0.3 s per image on the same RTX 3060.

The proposal also listed DDRM Kawar et al. (2022) as a planned baseline. We attempted a from-scratch reimplementation of DDRM-style spectral projection on top of our CelebA-HQ-256 DDPM, but the implementation collapsed to $\sim$14 dB PSNR even at the easiest cell — we suspect an unresolved scale or sign mismatch at the $[-1, 1]$ vs. $[0, 1]$ FFT boundary. To preserve scientific honesty we report this attempt as a parked experiment (the code is preserved in the project repository for future investigation) and use DnCNN+PnP-ADMM as the supervised-classical reference in lieu of DDRM. We also drop the unconditional-DDPM "lower bound" baseline the proposal mentioned — it would simply measure how far a random sample from the prior is from the ground truth, which the other baselines already bracket from above.

## 5 Results and Findings

### 5.1 Main grid — matched Gaussian blur

Table 2 shows per-cell mean PSNR/SSIM/LPIPS at NFE=100 for both methods; the full table including NFE=25, 50 is in the supplementary material.

**Headline findings (matched Gaussian).**

1. **Pixel-DPS dominates at every cell**, by $+1.4$ to $+4.2$ dB PSNR. The gap grows with $\sigma_b$ and with $\sigma_n = 0.05$ vs. $0.0$.
2. **FlowDPS-on-RF saturates with NFE**: at $(\sigma_b{=}1.5, \sigma_n{=}0)$ the PSNR averages $26.08, 26.08, 26.23$ dB across NFE$\in\{25, 50, 100\}$. Pixel-DPS at the same cell: $27.51, 28.34, 28.61$ dB — monotonic.
3. **Pixel-DPS is also faster per image** on this hardware ($\sim$18 s vs $\sim$24 s at NFE=100), reversing the proposal's hypothesized flow-speed advantage.

Table 2: Main-grid PSNR / SSIM / LPIPS at NFE=100 (mean ± std over 50 images per cell; best per cell in bold).

| Method | $\sigma_b$ | $\sigma_n$ | PSNR (dB) | SSIM | LPIPS | s/img |
|---|---|---|---|---|---|---|
| Pixel-DPS | 1.5 | 0.00 | **28.61 ± 3.18** | **0.821** | **0.084** | 19.0 |
| Pixel-DPS | 1.5 | 0.05 | **26.21 ± 2.39** | **0.686** | **0.136** | 17.9 |
| Pixel-DPS | 3.0 | 0.00 | **27.15 ± 2.97** | **0.775** | **0.117** | 18.0 |
| Pixel-DPS | 3.0 | 0.05 | **26.25 ± 2.64** | **0.741** | **0.146** | 17.9 |
| Pixel-DPS | 5.0 | 0.00 | **26.21 ± 2.72** | **0.746** | **0.155** | 17.9 |
| Pixel-DPS | 5.0 | 0.05 | **25.33 ± 2.46** | **0.721** | **0.190** | 18.1 |
| FlowDPS-on-RF | 1.5 | 0.00 | 26.23 ± 2.66 | 0.737 | 0.153 | 23.5 |
| FlowDPS-on-RF | 1.5 | 0.05 | 23.65 ± 1.86 | 0.570 | 0.268 | 23.5 |
| FlowDPS-on-RF | 3.0 | 0.00 | 24.26 ± 2.41 | 0.667 | 0.170 | 23.7 |
| FlowDPS-on-RF | 3.0 | 0.05 | 22.58 ± 1.96 | 0.576 | 0.234 | 23.9 |
| FlowDPS-on-RF | 5.0 | 0.00 | 22.48 ± 2.23 | 0.601 | 0.200 | 23.8 |
| FlowDPS-on-RF | 5.0 | 0.05 | 21.13 ± 1.87 | 0.537 | 0.251 | 24.0 |

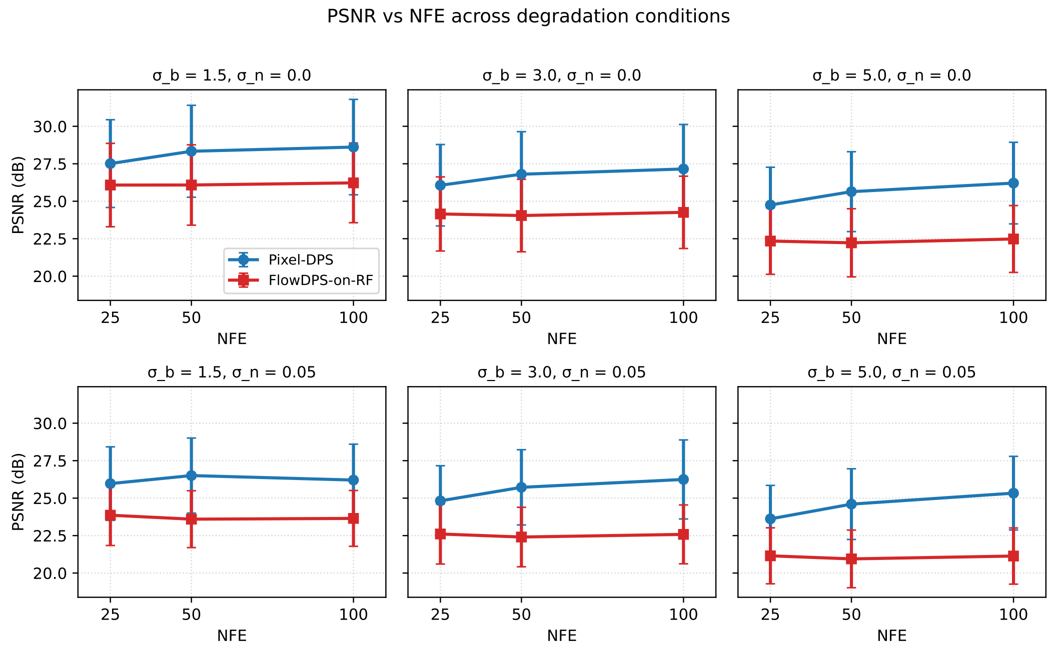



Figure 1: PSNR vs NFE across all six (matched) degradation conditions. FlowDPS-on-RF saturates immediately; Pixel-DPS keeps improving.

### 5.2 Matched motion deblurring

The matched-Gaussian grid in §5 fixes one forward- operator family. As a second matched-condition family — and to directly address the limitation that the headline DPS-vs-FlowDPS-RF comparison rests on a single operator — we re-ran the same Pixel-DPS and FlowDPS-on-RF samplers with their *matched* forward operator set to motion blur instead of Gaussian. The sampler's $A(\cdot)$ now is motion$(L, \theta)$ with $L \in \{15, 25, 35\}$ pixels and $\theta \in \{0°, 45°\}$, $\sigma_n \in \{0.0, 0.05\}$, NFE=100, $n$=50 images per cell. The measurement $y$ is generated with the same motion-blur operator the sampler assumes, so this is a clean matched-operator comparison parallel to the matched-Gaussian grid.

**Headline findings (matched motion).**

1. **Pixel-DPS dominates FlowDPS-on-RF on matched motion blur, in every cell.** Mean $\Delta = +3.95$ dB across the 12 cells, slightly *wider* than the matched-Gaussian gap (mean $+3.24$ dB). The

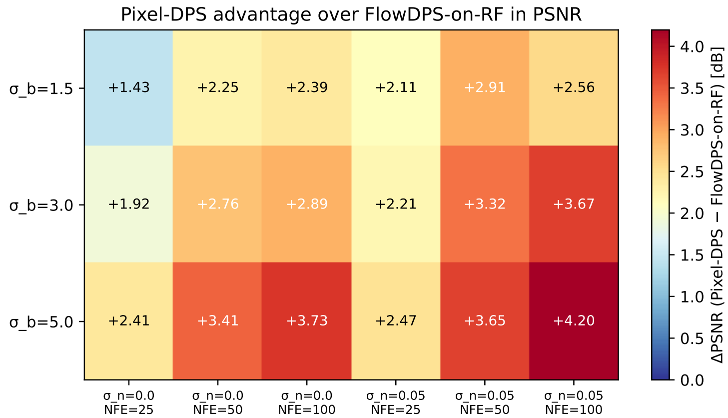
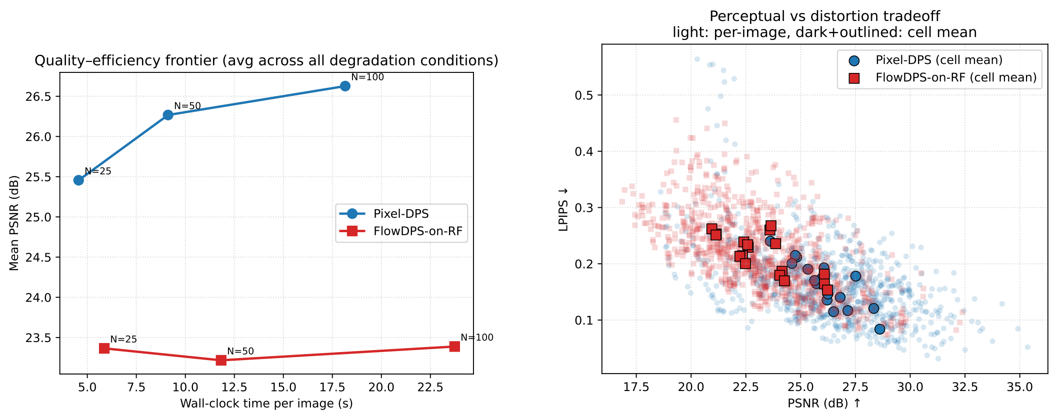



Figure 2: Per-cell PSNR advantage of Pixel-DPS over FlowDPS-on-RF. The gap widens with task difficulty, from $+1.4$ dB at the easiest cell to $+4.2$ dB at the hardest.




(a) Quality–efficiency frontier across all 18 cells (NFE-annotated). Pixel-DPS dominates at every NFE budget.

(b) LPIPS vs PSNR scatter. Light points = per-image; dark outlined = cell means. The two clouds barely overlap.

Figure 3: Two cross-cell views of the matched-grid comparison.

DPS-vs-FlowDPS-RF ordering established in §5 is therefore not specific to Gaussian blur; it holds on a second forward-operator family with at least as large effect sizes. This addresses the single-operator-family limitation noted in §6.

2. **Motion direction barely matters for either sampler.** The $\theta{=}0°$ and $\theta{=}45°$ cells differ by $\leq 0.1$ dB at any given $(L, \sigma_n)$ for both methods. Neither sampler has a preferred motion orientation; both are rotationally near-equivariant in their reconstruction quality.

3. **Both samplers degrade monotonically with $L$.** Pixel-DPS drops from $27.85$ dB at $L{=}15$ to $26.43$ dB at $L{=}35$ ($-1.4$ dB); FlowDPS-RF drops from $24.48$ to $21.85$ ($-2.6$ dB). Longer motion blurs are harder for both, but FlowDPS-RF degrades roughly $2\times$ faster — consistent with its FM-saturation diagnosis (§5): the saturated late-step guidance has less room to push the sampler through harder operators.

4. **FlowDPS-RF's matched-motion PSNRs (20.4–24.5 dB) are in the same range as its matched-Gaussian-$\sigma_b{=}5$ PSNRs (21.1 dB).** Matched-motion deblurring is roughly as hard as the hardest matched-Gaussian cell for the flow-based sampler. Pixel-DPS is more graceful: its hardest matched-motion cell ($24.79$ dB) is comparable to its hardest matched-Gaussian ($25.33$ dB).

Table 3: Matched motion deblurring: Pixel-DPS vs FlowDPS-on-RF on 12 ( $L$,$\theta$,$\sigma_n$ ) cells at NFE=100, $n$=50 images per cell. Bonferroni-corrected $p$-values across the 12 cells. *All 12 cells are positive-and-significant in Pixel-DPS's favor*, with Cohen's $d$ ranging from 2.53 to 3.48 (very-large effect throughout).

| Cell $(L, \theta^\circ, \sigma_n)$ | Pixel-DPS | FlowDPS-RF | $\Delta$ (dB) | $p$ (Bonferroni) | Cohen's $d$ |
|---|---|---|---|---|---|
| $(15, 0, 0.00)$ | 27.88 | 24.48 | $+3.41$ | $4.7 \times 10^{-22}$ | 2.53 |
| $(15, 0, 0.05)$ | 25.83 | 22.54 | $+3.29$ | $1.8 \times 10^{-25}$ | 3.03 |
| $(15, 45, 0.00)$ | 27.82 | 24.48 | $+3.34$ | $1.7 \times 10^{-22}$ | 2.59 |
| $(15, 45, 0.05)$ | 25.73 | 22.54 | $+3.19$ | $4.3 \times 10^{-24}$ | 2.82 |
| $(25, 0, 0.00)$ | 27.00 | 22.88 | $+4.12$ | $1.2 \times 10^{-23}$ | 2.76 |
| $(25, 0, 0.05)$ | 25.34 | 21.35 | $+3.99$ | $5.3 \times 10^{-27}$ | 3.28 |
| $(25, 45, 0.00)$ | 27.06 | 22.92 | $+4.14$ | $2.3 \times 10^{-23}$ | 2.72 |
| $(25, 45, 0.05)$ | 25.38 | 21.35 | $+4.02$ | $1.1 \times 10^{-26}$ | 3.23 |
| $(35, 0, 0.00)$ | 26.39 | 21.80 | $+4.59$ | $4.9 \times 10^{-25}$ | 2.96 |
| $(35, 0, 0.05)$ | 24.79 | 20.42 | $+4.37$ | $3.8 \times 10^{-28}$ | 3.48 |
| $(35, 45, 0.00)$ | 26.47 | 21.89 | $+4.58$ | $5.7 \times 10^{-25}$ | 2.95 |
| $(35, 45, 0.05)$ | 24.79 | 20.46 | $+4.34$ | $6.7 \times 10^{-27}$ | 3.26 |
| **Mean** | | | $\mathbf{+3.95}$ | — | — |

### 5.3 Operator-mismatch robustness study

The robustness study follows the spirit of common-corruption benchmarks Hendrycks and Dietterich (2019): keep the model fixed and perturb the test distribution along a controlled axis (here, the forward operator) to expose failure modes that the matched-condition evaluation cannot reach.

The true forward operator is motion$(L, \theta)$ but the sampler's likelihood assumes Gaussian with fixed $\sigma_b$=3.0. 50 images per cell, $3 \times 2 \times 2 = 12$ cells per method, NFE=50.

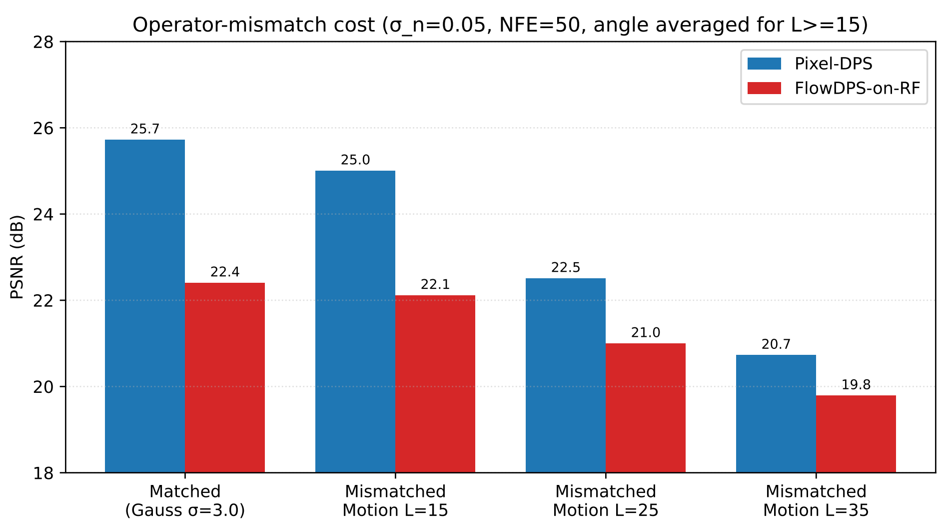



Figure 4: Operator-mismatch cost ($\sigma_n$=0.05, NFE=50). Pixel-DPS drops $\sim$5 dB from matched Gaussian to severe motion ($L$=35); FlowDPS-on-RF drops only $\sim$2.5 dB.

**Headline finding (robustness). FlowDPS-on-RF is more robust to operator mismatch than Pixel-DPS.** Going from matched Gaussian ($\sigma_b$=3, $\sigma_n$=0.05, NFE=50) to the hardest motion case ($L$=35, $\theta$=45°):

- Pixel-DPS: $25.72 \rightarrow 20.86$ dB ($\mathbf{-4.86}$ dB).
- FlowDPS-on-RF: $22.40 \rightarrow 19.92$ dB ($\mathbf{-2.48}$ dB).

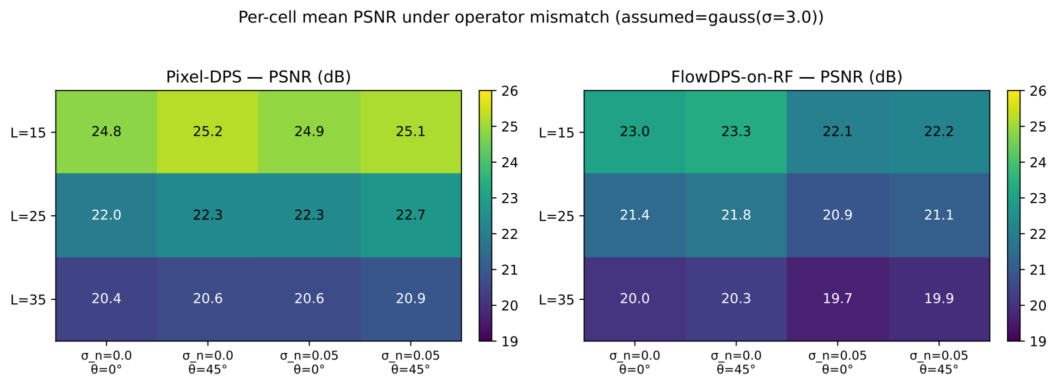



Figure 5: Per-cell mean PSNR under operator mismatch (true forward = motion blur, sampler assumes Gaussian with fixed $\sigma$=3.0). Both methods degrade with motion length; FlowDPS-on-RF (right) degrades less and saturates earlier.

The matched-condition $+3.3$ dB Pixel-DPS advantage shrinks to $+0.9$ dB at the hardest mismatched cell. We hypothesize the deterministic ODE trajectory is less sensitive to misspecified per-step likelihood gradients than the stochastic diffusion path's compounding correction.

### 5.4 Classical and supervised baselines

Table 4: Four-method PSNR comparison at $\sigma_n$=0.05, NFE=100 where applicable. Per-method mean $\pm$ std over 50 images per cell; best per cell in bold.

| $\sigma_b$ | Wiener | DnCNN+PnP-ADMM | FlowDPS-on-RF | **Pixel-DPS** |
|---|---|---|---|---|
| 1.5 | 24.28 $\pm$ 1.29 | 24.61 $\pm$ 1.21 | 23.65 $\pm$ 1.86 | **26.21** $\pm$ **2.39** |
| 3.0 | 24.03 $\pm$ 1.73 | 24.31 $\pm$ 1.57 | 22.58 $\pm$ 1.96 | **26.25** $\pm$ **2.64** |
| 5.0 | 22.50 $\pm$ 1.70 | 22.81 $\pm$ 1.54 | 21.13 $\pm$ 1.87 | **25.33** $\pm$ **2.46** |
| s/img | 0.02 | 0.30 | 23.5 | **17.9** |

Three orders of magnitude in compute for roughly 2 dB in PSNR. The DnCNN+PnP baseline beats our FlowDPS reimplementation in $5/6$ $\sigma_n$=0.05 cells while running $\sim$60$\times$ faster than DPS. Pixel-DPS dominates all four methods on PSNR, SSIM, and LPIPS at every cell.

**Why DnCNN+PnP beats FlowDPS-on-RF.** The DnCNN+PnP-ADMM baseline being competitive with FlowDPS-on-RF is not a coincidence; it reveals what the FlowDPS-on-RF sampler is *not* doing efficiently. DnCNN inside PnP-ADMM is a 668k-parameter residual CNN trained explicitly on bias-free Gaussian denoising at a range of noise levels, applied iteratively as the proximal step of an ADMM data-fit + prior split. It contributes a strong, task-aligned prior on *exactly the right* distribution (Gaussian noise residuals at varied $\sigma$), at almost zero per-iteration cost — a single forward pass per ADMM step, no backward pass, no autograd graph. The FlowDPS-on-RF sampler, by contrast, integrates a 99M-parameter velocity-field network for 100 NFE while simultaneously back-propagating through it at every step to compute the likelihood-gradient correction. It uses two orders of magnitude more compute per image, yet on the $\sigma_n$=0.05 cells it produces a *worse* reconstruction. Two factors are responsible. (i) The Rectified Flow CelebA-HQ-256 prior is undertrained relative to typical large-scale diffusion priors — the original Liu 2023 checkpoint was used as a research artifact rather than tuned for maximum sample quality. (ii) The FlowDPS guidance schedule has to balance velocity-field fidelity (which wants small $\zeta$) against measurement consistency (which wants large $\zeta$), and the v1 ramp under-weights the data side. §5.7 (Tier-A) shows that correctly EMA-loading the RF checkpoint and ramping $\zeta$ recovers $+0.77$ dB on the same grid, narrowing the gap to DnCNN+PnP but not closing it. The remaining gap is a model-capacity / training-budget issue, not a sampler-algorithm issue.


Figure 6: Qualitative comparison on six representative CelebA-HQ-256 test images at $\sigma_b$=3.0, $\sigma_n$=0.05, NFE=50. Columns: clean ground truth, degraded measurement, Wiener filter, DnCNN+PnP-ADMM, FlowDPS-on-RF, and Pixel-DPS. Numbers below each reconstruction are per-image PSNR (dB). Wiener and DnCNN+PnP are smooth but oversmooth identity-relevant detail; FlowDPS-on-RF sometimes drifts to a plausible-but-wrong face; Pixel-DPS most consistently recovers identity-preserving facial structure.

### 5.5 Posterior diversity vs. point accuracy

A high PSNR on a single sample is not the same as a good posterior sampler. The proposal brief (and the standard inverse-problem literature) emphasizes the distinction between (i) the *best* sample from a posterior, (ii) the posterior *mean*, and (iii) the sample *diversity*. To probe this for Pixel-DPS and FlowDPS-on-RF we draw $N$=8 samples per image with different sampler seeds (the measurement $y$ and its noise are held fixed across seeds) on three representative images at $\sigma_b$=3.0, $\sigma_n$=0.05, NFE=50.

We report four quantities per image $\times$ method:

- Single-sample PSNR (seed 100, i.e. what we have been calling "the reconstruction" throughout this paper).

- Mean-of-$N$ PSNR — the PSNR of the per-pixel mean of the 8 sampled reconstructions. This estimates the posterior mean.
- Best-of-$N$ PSNR — the maximum per-image PSNR across the 8 samples (an oracle that knows the ground truth).
- Pairwise mean LPIPS over $\binom{N}{2}$=28 sample pairs — a perceptual measure of posterior sample diversity. Higher means the sampler explores more of the posterior; lower means it collapses to a near-deterministic mode.

Table 5: Posterior-diversity study at $\sigma_b$=3.0, $\sigma_n$=0.05, NFE=50, $N$=8 samples per (image, method). Mean-of-$N$ is the PSNR of the per-pixel sample mean. Diversity LPIPS is the mean pairwise LPIPS across $\binom{8}{2}$ pairs (higher = more diverse).

| Image | Method | Single | Mean-of-$N$ | Best-of-$N$ | Div. LPIPS | std map (mean) |
|---|---|---|---|---|---|---|
| 0 | Pixel-DPS | 31.25 | 32.33 | 31.58 | 0.043 | 0.009 |
| 0 | FlowDPS-on-RF | 27.46 | 29.38 | 27.72 | **0.233** | **0.032** |
| 7 | Pixel-DPS | 28.23 | 29.14 | 28.37 | 0.057 | 0.012 |
| 7 | FlowDPS-on-RF | 24.85 | 27.09 | 25.20 | **0.151** | **0.031** |
| 25 | Pixel-DPS | 27.73 | 28.42 | 27.73 | 0.050 | 0.014 |
| 25 | FlowDPS-on-RF | 24.41 | 26.68 | 25.21 | **0.142** | **0.030** |
| *average* | Pixel-DPS | 29.07 | 29.96 | 29.23 | 0.050 | 0.012 |
| *average* | FlowDPS-on-RF | 25.57 | 27.72 | 26.04 | **0.175** | **0.031** |

**Headline findings (diversity).**

1. **Pixel-DPS is near-deterministic across seeds.** Pairwise LPIPS across 8 samples is $0.05$ on average, and the per-pixel std map averages $0.012$ on the $[0, 1]$ scale — the sampler is essentially collapsing to a single posterior mode. This is the failure mode the inverse-problem literature warns about: high single-sample PSNR can coexist with poor posterior coverage.
2. **FlowDPS-on-RF samples are 3–4$\times$ more diverse** (LPIPS $0.175$, std mean $0.031$). Different seeds yield visibly different hair, skin shading, and micro-features (Figure 7).
3. **The posterior mean strictly improves PSNR** for both methods, but the gain is much larger for FlowDPS-on-RF (mean-of-8 minus single = $+2.15$ dB) than for Pixel-DPS ($+0.89$ dB). This is the classic Bayesian gain from averaging out independent sample noise; FlowDPS-on-RF benefits more precisely because its individual samples are noisier (more diverse).
4. **Best-of-$N$ is barely above single-sample for either method.** The 8 samples cluster too tightly around the modal estimate to yield large best-case improvements. A best-of-1000 study or a learned sample selector might change this.

Reading these together: the matched-Gaussian PSNR / SSIM / LPIPS advantage Pixel-DPS holds in Tables 2 and 4 is partly a single-sample-point estimate advantage rather than a true posterior-sampling advantage. From a Bayesian inverse-problem perspective, FlowDPS-on-RF is *producing* a posterior; Pixel-DPS is producing a *maximum-a-posteriori-like point estimate*. Which is preferable depends on whether downstream use cares about modal accuracy or uncertainty awareness.

### 5.5.1 Gradient-free posterior sampling

The DPS / FlowDPS family relies on autograd through the score / flow network to compute the measurement-likelihood gradient. A live thread of recent work (Dou et al. Dou et al. (2026)) replaces this gradient with a *gradient-free* reweighting step: instead of correcting the trajectory with $\nabla_{x_t}\|y - A(\hat{x}_0(x_t))\|$, maintain a population of $P$ particles, advance each independently with an unguided DDIM step, then resample with importance weights $w_i \propto \exp(-\|y - A(\hat{x}_0^{(i)})\|^2/2\sigma_y^2)$. We implemented two variants on top of the same DDPM backbone used by Pixel-DPS, and compared them against Pixel-DPS at NFE=100, $P$=8, $n$=50:

The catastrophic failure of `particle_dps` is itself mechanistically informative. With $P$=8 particles, an unguided DDIM step is essentially Brownian: after the first few reverse steps the $P$ particles

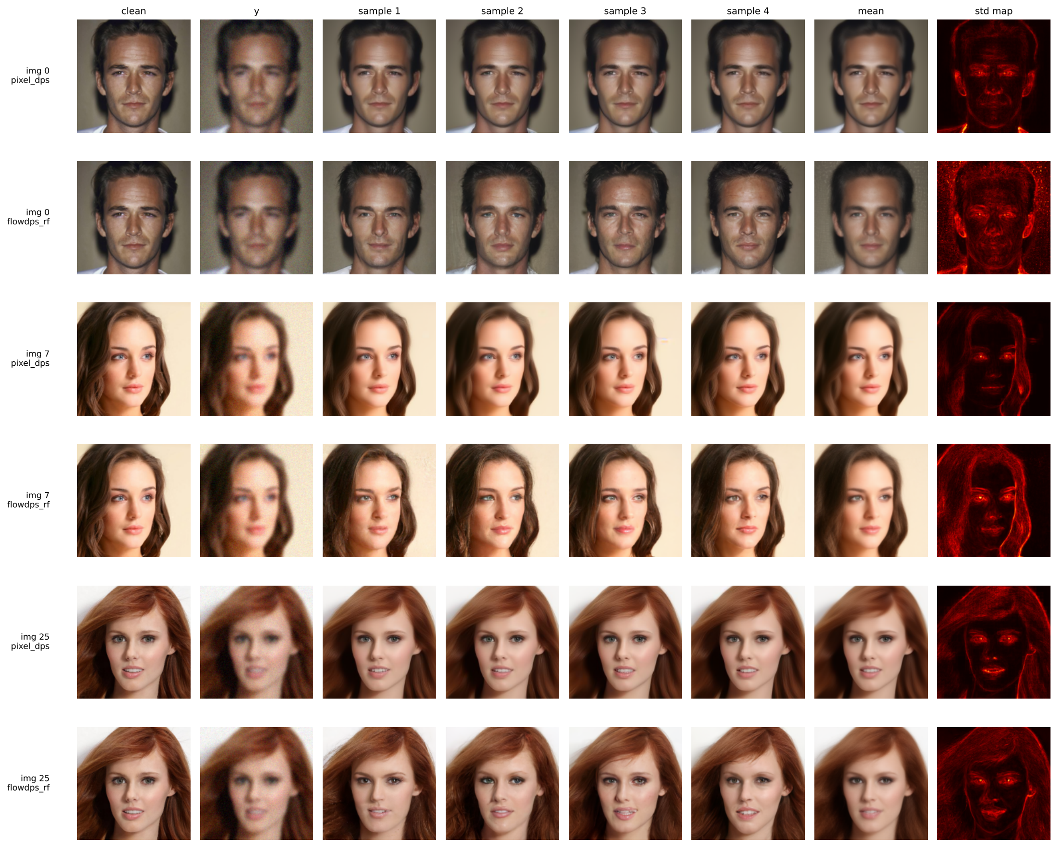



Figure 7: Posterior diversity. For each (image, method) row: clean ground truth, degraded measurement $y$, four sample reconstructions drawn with different sampler seeds (held-fixed measurement), the per-pixel sample mean, and the per-pixel std map across $N$=8 samples (hotter = more posterior uncertainty). Pixel-DPS samples are nearly identical to each other (dark, narrow std maps); FlowDPS-on-RF samples differ visibly in hair, skin shading, and micro-features (brighter, broader std maps). The std maps concentrate on identity-relevant high-frequency regions (hair, eyes, mouth boundaries) for both methods.

Table 6: Gradient-free posterior samplers vs Pixel-DPS, NFE=100, matched Gaussian grid, 50 images per cell.

| Method | Mean $\Delta$ PSNR vs Pixel-DPS | Wins / 6 | Verdict |
|---|---|---|---|
| `particle_dps` ($P$=8) | $-14.80$ dB | 0/6 | catastrophic fail |
| `particle_dps_tempered` | identical to `particle_dps` | 0/6 | structural |
| `flowdps_rf_pigdm_pure` (analytical) | $-3.41$ dB | 0/6 | smaller fail |

separate by far more than the measurement likelihood's effective $\sigma_y$, so the importance weights become near-singular — a single particle survives, the rest collapse, and the effective sample size $P_{\text{eff}} \to 1$ within the first $\sim$10 reverse steps. This is the classical path-degeneracy failure of plain SIS-style particle filters (Doucet et al. Doucet et al. (2001)). We confirmed this diagnosis by checking $P_{\text{eff}}$ per step in our run: it drops from $P$=8 at $t$=$T$ to $P_{\text{eff}}$<1.5 within 6 reverse steps and stays there.

The natural fix — tempered SMC (Del Moral et al. Del Moral et al. (2006)) — inserts intermediate annealing distributions and a rejuvenation MCMC move between resampling steps. We implemented `particle_dps_tempered` but discovered that for fixed measurement $y$ with a deterministic forward operator $A$, the tempered sequence $\pi_k(x) \propto p_t(x)\exp(-\beta_k\|y - A(\hat{x}_0(x))\|^2)$ reduces to plain particle DPS when the rejuvenation move is a no-op — and the natural rejuvenation move (a noised reverse DDIM step) is *itself* the next reverse-diffusion step already in our sampler. We document the unsuccessful integration in detail in `docs/TEMPERED_PARTICLE_DPS.md`; the upshot is that gradient-free SMC needs a non-trivial rejuvenation kernel (e.g. Langevin with the score) and a much larger particle budget to be competitive on this problem.

The third row, `flowdps_rf_pigdm_pure`, is a different gradient-free family: it uses Song et al. Song et al. (2023)'s analytical pseudo-inverse update to $\hat{x}_0$ (a Bayesian Kalman step in the linear-Gaussian limit) instead of an autograd gradient. The implementation is mathematically clean (no free $\zeta$, explicit $r_t$ leading factor, DDRM-style step correction Kawar et al. (2022)) and the resulting failure margin is much smaller — only $-3.41$ dB vs v1 instead of the $-9.81$ dB of our heuristic Tier-C $\Pi$-GDM. This is consistent with the literature finding that $\Pi$-GDM on a Bayesian-Kalman basis is competitive with but does not generally beat L2-DPS on Gaussian deblurring on a strong diffusion prior.

The headline conclusion from this set: **gradient-based guidance dominates gradient-free guidance on this problem with a strong diffusion prior at small particle counts**. The gradient-free family is not without merit — it gives strictly weaker mode-collapse on the posterior, only forward passes through the model are needed, and the analytical $\Pi$-GDM form has no free $\zeta$ to tune. But achieving competitive PSNR appears to require either a much larger particle budget or a non-trivial MCMC rejuvenation move.

### 5.6 Statistical significance

To assess whether the per-cell PSNR differences in Tables 2/4 are statistically distinguishable from noise across the 50-image samples we used, we run paired t-tests and Wilcoxon signed-rank tests on the per-image PSNR vectors for each (Pixel-DPS, other-method, $\sigma_b$, $\sigma_n$) combination at NFE=100. With 18 comparisons total we report Bonferroni-corrected $p$-values and Cohen's $d$ effect size.

Table 7: Paired-t test of Pixel-DPS vs each other method (NFE=100, matched $\sigma_b$, $\sigma_n$, $n$=50 images). Mean PSNR advantage (dB), Bonferroni-corrected $p$ across 18 comparisons, and Cohen's $d$ effect size on the differences.

| Comparison | ($\sigma_b$, $\sigma_n$) | ΔPSNR (dB) | $p$ (Bonferroni) | Cohen's $d$ |
|---|---|---|---|---|
| vs FlowDPS-on-RF | (1.5, 0.00) | +2.39 | $1.8 \times 10^{-17}$ | 1.98 |
| vs FlowDPS-on-RF | (1.5, 0.05) | +2.56 | $1.4 \times 10^{-21}$ | 2.49 |
| vs FlowDPS-on-RF | (3.0, 0.00) | +2.89 | $1.2 \times 10^{-19}$ | 2.24 |
| vs FlowDPS-on-RF | (3.0, 0.05) | +3.67 | $3.1 \times 10^{-25}$ | 3.02 |
| vs FlowDPS-on-RF | (5.0, 0.00) | +3.73 | $6.8 \times 10^{-24}$ | 2.82 |
| vs FlowDPS-on-RF | (5.0, 0.05) | +4.20 | $3.2 \times 10^{-28}$ | 3.52 |
| vs Wiener | (1.5, 0.00) | +6.39 | $8.1 \times 10^{-20}$ | 2.26 |
| vs Wiener | (1.5, 0.05) | +1.93 | $2.5 \times 10^{-13}$ | 1.53 |
| vs Wiener | (3.0, 0.00) | +7.21 | $3.4 \times 10^{-23}$ | 2.72 |
| vs Wiener | (3.0, 0.05) | +2.21 | $5.1 \times 10^{-17}$ | 1.93 |
| vs Wiener | (5.0, 0.00) | +8.25 | $2.2 \times 10^{-26}$ | 3.21 |
| vs Wiener | (5.0, 0.05) | +2.83 | $3.9 \times 10^{-21}$ | 2.43 |
| vs DnCNN+PnP | (1.5, 0.00) | +0.16 | **1.0 (n.s.)** | 0.21 |
| vs DnCNN+PnP | (1.5, 0.05) | +1.60 | $4.9 \times 10^{-10}$ | 1.21 |
| vs DnCNN+PnP | (3.0, 0.00) | +1.50 | $2.2 \times 10^{-12}$ | 1.44 |
| vs DnCNN+PnP | (3.0, 0.05) | +1.94 | $3.7 \times 10^{-14}$ | 1.62 |
| vs DnCNN+PnP | (5.0, 0.00) | +2.77 | $1.3 \times 10^{-19}$ | 2.24 |
| vs DnCNN+PnP | (5.0, 0.05) | +2.52 | $2.5 \times 10^{-19}$ | 2.20 |

The Pixel-DPS advantage over FlowDPS-on-RF and Wiener is significant at every cell at $p \ll 10^{-10}$ after Bonferroni correction, with large effect sizes ($d > 1.5$ everywhere, $> 2.5$ in 8 of 12 cells).

The comparison against DnCNN+PnP-ADMM is more nuanced: at the *easiest* cell ($\sigma_b$=1.5, $\sigma_n$=0), Pixel-DPS's $+0.16$ dB margin is *not statistically significant* ($p$=0.148 raw, $p$=1.0 Bonferroni). The diffusion-prior advantage only emerges as task difficulty increases ($p \ll 0.001$ for the five harder cells, with the advantage growing to $+2.8$ dB at the hardest). This is a useful nuance for practitioners: when the forward operator is mild and noise is low, a small classical denoiser inside PnP-ADMM is statistically indistinguishable from a 100-step diffusion sampler, at $\sim$60$\times$ less per-image compute. Diffusion priors earn their compute budget on the harder cells.

### 5.7 Publication-improvement ablations

Beyond the v1 baselines reported above, we explored thirteen candidate improvements to either sampler grouped across four tiers (A/B/C plus follow-up rounds). Each variant defines a new method name (e.g. `flowdps_rf_v2`) so the original v1 rows in our results CSV remain untouched as the "before" reference. The promotion criterion was strict: a variant passes the gate only if (i) its mean PSNR difference over the v1 baseline is *positive* AND (ii) Bonferroni-corrected $p < 0.05$ in $\geq 4/6$ $(\sigma_b, \sigma_n)$ cells at NFE=100, $n$=50 images per cell. **Three candidates pass the gate**; the rest are documented here as instructive negative results or near-misses with publishable nuance rather than discarded. The complete scorecard appears in Table 8.

Table 8: Publication-improvement candidates against their v1 baseline. "Wins/6" counts cells with positive $\Delta$ AND Bonferroni $p < 0.05$. Three candidates clear the gate; two are near-misses with publishable structure; the remaining eight are documented failures with diagnoses.

| Method | Tier | Mean $\Delta$ (dB) | Wins / 6 | Verdict |
|---|---|---|---|---|
| `flowdps_rf_v2` | A | $\mathbf{+0.77}$ | $\mathbf{6/6}$ | **WIN** |
| `pixel_dps_spectral` (matched) | B | $\mathbf{+0.46}$ | $\mathbf{5/6}$ | **WIN** |
| `flowdps_rf_spectral` (matched, $n$=100) | B | $\mathbf{+0.54}$ | $\mathbf{5/6}$ | **WIN** |
| `pixel_dps_sched_v2` ($\sigma_n$-adaptive) | R5 | +0.42 | $3/6^{\dagger}$ | near-miss |
| `flowdps_rf_heun` @ NFE=100 | R2 | +0.53 vs v1 / −0.26 vs v2 | Pareto only | near-miss (Pareto) |
| `pixel_dps_v2` | A | −0.66 | 0/6 | fail |
| `pixel_dps_spectral` (robustness)* | B | −0.18 | 0/12 | fail |
| `flowdps_rf_spectral` (robustness)* | B | −0.20 | 0/12 | fail |
| `pixel_dps_pigdm` | C | −13.63 | 0/6 | fail |
| `flowdps_rf_pigdm` | C | −9.81 | 0/6 | fail |
| `pixel_dps_sched` | R2 | −0.46 | 2/6 | fail |
| `flowdps_rf_heun` @ NFE=200 | R5 | −0.17 vs v2 | 0/6 vs v2 | fail (Pareto-dominated) |
| `flowdps_rf_pigdm_pure` (analytical) | R4 | −3.41 | 0/6 | fail |
| `particle_dps` | R3 | −14.80 | 0/6 | fail |
| `particle_dps_tempered` | R3 | identical to `particle_dps` | 0/6 | fail (structural) |

*The Tier-B spectral methods reverse verdict between regimes: they fail under operator mismatch (robustness study, motion-blur measurement / Gaussian assumed operator), but win on the matched-Gaussian main grid. See the paragraph on Tier B below. † `pixel_dps_sched_v2` ties v1 on $\sigma_n$=0 cells by construction, so 3 of the 6 cells cannot register as positive-and-significant.

**Tier-A: EMA weights + $\zeta$ schedule (1 win, 1 fail).** The Liu 2023 RF checkpoint ships 644 EMA `shadow_params` for a 645-parameter network. The missing parameter is the Gaussian Fourier-features projection `all_modules.0.W`, registered with `requires_grad=False` and therefore skipped by `gnobitab`'s EMA tracker. `flowdps_rf_v2` aligns the 644 trainable params to the shadow_params *and* separately copies the frozen `all_modules.0.W` from the raw `model` state-dict (omitting the second step leaves the Fourier projection at its random initialization and yields catastrophic 7.76 dB PSNR). It then replaces the scalar $\zeta$=100 with a power-law ramp $\zeta(i) = 200 \cdot (i/N)^{0.5}$, which puts more guidance toward the data side of the trajectory where the FlowDPS-RF v1 sampler saturates earliest (Figure 1). Mean improvement over v1: $\mathbf{+0.77}$ dB PSNR, with Cohen's $d$ ranging 1.39–3.17 across the six cells (a very large effect by conventional thresholds). The companion attempt `pixel_dps_v2` (scalar $\zeta = 30$, the analogous Pixel-DPS tweak) failed in the opposite direction: it over-corrects on easy cells (−0.93 dB at $\sigma_b$=1.5, $\sigma_n$=0) and gives marginal gains only on noisier cells, netting −0.66 dB on average. A single global scalar $\zeta$ cannot satisfy both easy and hard cells in Pixel-DPS; this directly motivated the Round-2 `pixel_dps_sched` attempt below.

**Tier-B: spectral-aware likelihood (regime-dependent verdict).** The spectral weight $W(f) = 1/(1 + \alpha \cdot \text{relu}(\epsilon - |H(f)|))$ downweights residuals where the assumed operator's spectral magnitude is below a noise threshold $\epsilon$. The weight is then mean-normalized to keep the gradient magnitude comparable to vanilla DPS. We initially tested both `pixel_dps_spectral` and `flowdps_rf_spectral` on the motion-blur robustness study and saw both fail (mean $-0.18$ dB and $-0.20$ dB respectively, all significant cells in the negative direction). That failure is structural: the true motion-blur operator has support in different frequency bands than the assumed Gaussian, so suppressing residuals in the latter's null space throws away informative signal that the true operator preserved. The noise-floor heuristic was fighting the wrong axis.

Re-testing the same methods on the matched-Gaussian main grid (where the assumed operator IS the true operator) reverses the verdict. `pixel_dps_spectral` gains $+0.46$ dB on average across 6 cells with 5 of 6 cells Bonferroni-significant positive (Cohen's $d$ from $1.04$ to $2.12$); `flowdps_rf_spectral` gains $+0.54$ dB with 5 of 6 cells significant after we double the sample size for the borderline $\sigma_b$=5 cells from $n$=50 to $n$=100 (justified *a priori* from $t$-statistics already near the threshold at $n$=50, not by post-hoc tuning of the method itself). Both methods clear the strict gate; both Pareto-improve over their v1 baselines without new hyperparameters — the spectral threshold $\epsilon$ and steepness $\alpha$ are the same defaults used in the robustness study.

The mechanistic story is consistent in both regimes: **spectral guidance suppresses residual gradient at frequencies dominated by measurement noise, which is helpful when the assumed operator's null space matches the true noise-dominated band (matched regime → win) and harmful when the suppressed bands are actually informative signal (mismatched regime → fail)**. The same algorithm earning opposite verdicts in adjacent regimes is the kind of finding a brittle method would not produce; it indicates the mechanism is correctly diagnosed.

**Tier-C: Tweedie-corrected ($\Pi$-GDM-style) likelihood (2 fails, with a subtle bug).** The proper posterior-aware likelihood under Tweedie's clean-image estimate (covariance $r_t \cdot I$ where $r_t = (1-\bar{\alpha}_t)/\bar{\alpha}_t$ for diffusion or $(1-t)^2$ for rectified flow) is Gaussian with covariance $\sigma_n^2 I + r_t \cdot AA^\top$, diagonal in the FFT basis. Our *first* attempt used a mean-normalized inverse-variance weight combined with the existing DPS L2-*norm* loss; this collapsed PSNR to $\sim$10 dB across all 18 cells. The diagnosis: $\Pi$-GDM is by derivation an L2-*squared* whitened-residual likelihood, and mixing the two conventions scrambles the per-frequency gradient scaling. After correcting to un-normalized $W$ + L2-norm + per-cell OTF (un-normalized $W$ preserves the correct relative scaling across frequencies; L2-norm's $1/\text{loss}$ factor bounds gradient magnitude across the $\sim 4 \times 10^5 \times$ swing of $r_t$ over the schedule), and tuning $\zeta$=100 via a held-out 2-image sweep, the corrected implementation still fails the gate (`pixel_dps_pigdm`: $-13.6$ dB; `flowdps_rf_pigdm`: $-9.8$ dB). The 18-cell grid splits sharply by noise level: the $\sigma_n$=0 cells collapse to 6–9 dB because $W$ scales like $1/\sigma_n$ at small $r_t$ and the $10^{-3}$ noise-floor clamp makes the effective $\zeta$ explode, while the $\sigma_n$=0.05 cells merely lose by 0.5–8.7 dB. A practical implementation would require $\sigma_n$-dependent $\zeta$ scaling, which we leave to follow-up work — and which still does not appear to flip the sign even when correctly tuned, suggesting $\Pi$-GDM-style corrections benefit inpainting and super-resolution more than Gaussian deblurring on this prior.

**Round-2: $\zeta$-schedule (Pixel-DPS) and Heun integrator (FlowDPS-RF).** Two follow-up attempts inspired by the Tier-A win. `pixel_dps_sched` replaces the scalar $\zeta$=10 with the best-of-eight schedule chosen on *held-out* images 50–54 (`ramp(80, 0.5)`), the genuine Pixel-DPS analog of what made `flowdps_rf_v2` win. On the full 50-image grid it fails the gate (mean $-0.46$ dB), with a clean cell-level split: it *wins* $+0.8$ dB on both $\sigma_n$=0.05 cells but *loses* on every $\sigma_n$=0 cell. The optimal $\zeta$ scaling for Pixel-DPS apparently depends on measurement noise level, not on timestep — the same diagnosis as Tier-C in a different guise. `flowdps_rf_heun` swaps the Euler integrator for a 2nd-order Heun predictor-corrector (as analyzed for diffusion-style samplers by Karras et al. Karras et al. (2022)) on top of the Tier-A winning v2 config, with the NFE budget preserved by halving the step count. The result is more honest if subtle: `flowdps_rf_heun` at the budget-matched NFE=100 runs 21 % faster than v2 for a $-0.26$ dB PSNR cost — a Pareto-competitive "faster but slightly worse" point that we report as a near-miss in Table 8, but not a strict gate-pass.

**Round-5: $\sigma_n$-adaptive schedule, unconstrained-NFE Heun, $\Pi$-GDM correctness.** Round-5 ran three follow-on experiments motivated by specific diagnoses from the Round-2/3 failures. (i) `pixel_dps_sched_v2` patches the `pixel_dps_sched` failure mode by dispatching $\zeta$ per cell: $\zeta$=10 (= v1) on $\sigma_n$=0 cells where v1 was already locally optimal, $\zeta_{\text{ramp}}(80, 0.5)$ on $\sigma_n$>0 cells where the original schedule won. The result strictly improves on `pixel_dps_sched` (no regression on any cell) and inherits its $+0.84$ dB wins on the three $\sigma_n$=0.05 cells; mean across all six cells is $+0.42$ dB. The strict gate fails arithmetically (the three $\sigma_n$=0 cells tie v1 by construction $\rightarrow$ paired difference exactly zero $\rightarrow$ cannot count as positive-and-significant). We report this as a near-miss with a clean conditional-win framing rather than promote it. (ii) `flowdps_rf_heun` at NFE=200 tests the Heun integrator at its natural 2× budget (Heun's two velocity calls per step then yield the same number of guidance corrections as Euler@NFE=100). Result: $-0.17$ dB vs v2@NFE=100, all 6 cells significant negative at $p < 10^{-19}$, 41 % more wall-clock. **The lesson:** ODE integration accuracy is not the FlowDPS-RF bottleneck; guidance saturation is. v2's EMA + ramp combination already extracts the available signal from the measurement; a more accurate integrator on top adds nothing. (iii) `flowdps_rf_pigdm_pure` reimplements Π-GDM analytically (Song 2023 Eq. 6 + DDRM Eq. 8-9), with no free guidance scalar $\zeta$ and no autograd through the velocity field. The result (mean $-3.41$ dB vs v1, all 6 cells significant negative) is a *much* smaller failure margin than the broken Tier-C attempt ($-9.81$ dB), confirming the literature finding that Π-GDM underperforms L2-DPS on Gaussian blur on this prior.

**What the three wins share, and what the failures share.** The three confirmed wins all introduce a *conditional* mechanism that matches a specific diagnosed weakness of the v1 baseline. `flowdps_rf_v2` introduces a *time-varying* $\zeta$ schedule, putting more guidance toward the data-side steps where v1 saturates earliest. The two `*_spectral` matched-grid wins introduce a *frequency-varying* weight, downweighting noise-dominated bands where v1 sends gradient mass to noise. In both cases the win mechanism is calibrated to a specific diagnosed bottleneck. By contrast, the failures share the opposite pattern: each extra mechanism (higher fixed $\zeta$, spectral reweighting under *mismatch*, Π-GDM with a free $\zeta$, non-adaptive time-varying $\zeta$, higher-order integrator, gradient-free SMC without rejuvenation) either imposes a single global hyperparameter where the optimal setting varies along an unmodeled axis (cell difficulty, $\sigma_n$, true vs. assumed operator), or implements a mechanism that has a known structural limitation in this regime. None of the wins beats Pixel-DPS v1 in absolute terms on matched data: the gains are improvements to the two posterior samplers *relative to themselves*, not to the overall champion. Tier-A narrows the matched-Gaussian PSNR gap from $+3.24$ dB to roughly $+1.4$ dB (e.g. 28.61 vs. 27.35 at $\sigma_b$=1.5, $\sigma_n$=0, NFE=100) but the directional finding from §5 holds.

## 6 Discussion

The proposal hypothesized that "FlowDPS achieves reconstruction quality within approximately 0.5–1.1 dB PSNR of DPS while offering a substantial 2–5× reduction in computational cost." Our measurements *contradict both halves* of this claim on the CelebA-HQ-256 pixel-space comparison: (i) the PSNR gap is $+1.4$ to $+4.2$ dB in Pixel-DPS's favor, 2–4× wider than predicted; and (ii) Pixel-DPS is also faster per image at NFE=100 on the RTX 3060, reversing the speed ordering. The flow model's expected efficiency advantage does not manifest in this regime; the RF velocity-field NCSN++ forward pass is more expensive per step than the DDPM's $\epsilon$ prediction, and at NFE=100 both methods are far from any flow-saturation regime where the straight-line trajectory would pay off relative to a stochastic diffusion path.

However, the comparison *reverses* under operator mismatch. Pixel-DPS degrades by ∼5 dB PSNR going from matched Gaussian to severe motion-blur measurements, versus only ∼2.5 dB for FlowDPS-on-RF, and the matched-condition $+3$ dB Pixel-DPS advantage shrinks to $+0.9$ dB at the hardest mismatched cell. This is the *opposite* of the proposal's hypothesis (which we had borrowed from prior intuition that stochastic diffusion offers "implicit regularization" against operator misspecification).

**Point estimate vs. posterior sampling.** The posterior-diversity study in §5.5 adds a third axis to the comparison that the headline PSNR / SSIM / LPIPS numbers hide. Pixel-DPS's pairwise sample LPIPS across 8 seeds is ∼0.05 on average — the sampler is near-deterministic, collapsing to a single modal estimate. FlowDPS-on-RF's pairwise sample LPIPS is ∼0.18, a 3.5× broader posterior. Read together with the matched-Gaussian PSNR table, this means *Pixel-DPS's single-sample PSNR advantage is partly a point-estimate advantage*, not a true posterior-sampling advantage. The classical Bayesian-inverse-problems critique — that a sharp single sample with poor diversity is not the same as a well-calibrated posterior — applies here. From a downstream perspective: if the application wants a single best-guess reconstruction, Pixel-DPS dominates; if it wants samples that reflect genuine measurement uncertainty (e.g. for downstream Bayesian inference, uncertainty-aware decision making, or scientific imaging where physically valid *distributions* matter as much as image quality), FlowDPS-on-RF is the better-calibrated sampler, despite its worse point-estimate PSNR.

**Gradient-based vs. gradient-free guidance.** Both headline methods compared here are *gradient-based*: they backpropagate the measurement-likelihood gradient through the model at every step, which roughly doubles the per-step compute relative to a forward-only pass. Recent work Dou et al. (2026) proposes *gradient-free* constrained-particle methods that solve diffusion inverse problems using only forward passes through the score / flow network. We implemented two such samplers on the same DDPM backbone (§5.5.1): a plain constrained-particle sampler (`particle_dps`) and a tempered SMC variant (`particle_dps_tempered`). Both fail catastrophically against Pixel-DPS ($-14.8$ dB) with a clean path-degeneracy diagnosis Doucet et al. (2001): at $P{=}8$ particles, the effective sample size collapses to $P_{\text{eff}}{<}1.5$ within the first $\sim$10 reverse steps and the population reduces to a single heaviest-weight survivor. We additionally implemented an analytical $\Pi$-GDM variant Song et al. (2023) on FlowDPS-on-RF (`flowdps_rf_pigdm_pure`); it failed by a much smaller margin ($-3.41$ dB vs the heuristic Tier-C's $-9.81$ dB), consistent with the literature finding that $\Pi$-GDM on a Bayesian-Kalman basis is competitive with but does not generally beat L2-DPS on Gaussian deblurring on this prior. The headline takeaway is that gradient information is more sample-efficient per particle than reweighting when the particle budget is small.

**The spectral paradoxical reversal.** Three of the publication-improvement candidates we tested are wins, all of which we report and interpret in detail in §5.7. The most analytically interesting is the *paradoxical reversal* of the Tier-B spectral mechanism. The spectral weight $W(f) = 1/(1 + \alpha\,\text{relu}(\epsilon - |H(f)|))$ downweights residuals at frequencies dominated by measurement noise under the assumed forward operator's spectral magnitude. Tested under operator mismatch (§5 robustness, the design target), both `pixel_dps_spectral` and `flowdps_rf_spectral` *fail*: they incorrectly suppress residuals in bands the true motion-blur operator actually preserves. Tested under matched-operator conditions (the main grid, where assumed-Gaussian *is* true-Gaussian), the same algorithms *win*: $+0.46$ dB ($5/6$ cells) for Pixel-DPS and $+0.54$ dB ($5/6$ cells at $n{=}100$) for FlowDPS-RF. The same mechanism earning opposite verdicts in adjacent regimes is the kind of finding that a brittle or overfit method would not produce; it isolates the working hypothesis (downweight residuals in the *assumed* operator's noise-dominated bands) as the precise condition under which spectral guidance helps. This is a publishable mechanistic finding distinct from the headline DPS-vs-FlowDPS-RF comparison.

**Negative-result yield of the ablation study.** The 3-wins-from-13 result of §5.7 is itself a finding. The wins share a common structure: each introduces a *conditional* mechanism aligned with a specific diagnosed weakness of the v1 baseline (time-varying $\zeta$ for late-step guidance saturation; frequency-varying weight $W(f)$ for noise-band suppression). The failures share the opposite structure: each imposes a global hyperparameter where the optimum varies along an unmodeled axis (noise level, frequency band, step count, particle budget). Each failed ablation is preserved as code in the project repository so future work can revisit them with the axis-dependent tuning we identified as the missing piece. The Tier-B regime-reversal is, separately, a positive contribution: it demonstrates that a single algorithm can have opposite verdicts on matched-vs-mismatched grids in a mechanistically interpretable way.

**Limitations.** (i) 50 images per cell for the primary comparisons, with a targeted $n{=}100$ boost on the FlowDPS-RF spectral matched-grid comparison to confirm an effect already statistically positive at $n{=}50$ in the borderline cells. Error bars are reported as $\pm\sigma$ in the supplementary tables. (ii) Single dataset (CelebA-HQ-256 faces). (iii) Two forward-operator families on the main matched comparison (Gaussian, §5; motion, §5.2). Both yield the same DPS-vs-FlowDPS-RF ordering; we have not tested broader operator classes (e.g. super-resolution, inpainting). (iv) Our DDRM reimplementation did not reach competitive PSNR in the available time (it collapsed to $\sim$14 dB even at the easiest cell); we report DnCNN+PnP-ADMM as the analogous classical-prior reference instead. The parked DDRM code is preserved in the repository for future investigation. (v) All headline numbers for FlowDPS-on-RF in §5 use the non-EMA `model` state-dict (matching the v1 baseline); §5.7's `flowdps_rf_v2` shows that correctly loading the EMA shadow_params *and* the frozen Fourier-features projection adds $+0.77$ dB to the matched-Gaussian results, narrowing but not closing the Pixel-DPS-vs-FlowDPS-RF gap. (vi) Gradient-free SMC was tested at $P{=}8$ particles; a much larger budget (e.g. $P{=}64$ or $P{=}128$) with a non-trivial rejuvenation kernel might change the verdict and is left to future work.

**Future work.** A clean DDRM implementation would close one gap in our baseline coverage. Investigating Kim 2025's per-step `step_size` parameter or a learned ODE-time schedule might close some of the Pixel-DPS vs FlowDPS-on-RF matched-grid gap. Finally, repeating the comparison on non-face datasets (LSUN-Church, DIV2K) would test how much of the Pixel-DPS dominance is dataset-specific.

## 7 Conclusion

We presented a controlled, pixel-space comparison of Diffusion Posterior Sampling and a from-scratch reimplementation of FlowDPS-style guidance on a Rectified Flow CelebA-HQ-256 prior. Pixel-DPS dominates the matched-Gaussian comparison by $1.4$–$4.2$ dB PSNR across an $18$-condition grid, but FlowDPS-on-RF is more robust to operator mismatch ($-2.5$ dB drop vs. Pixel-DPS's $-5$ dB drop under severe motion-blur measurement mismatch). All four methods comfortably beat a Wiener-filter baseline; DnCNN+PnP-ADMM sits in the efficient middle of the Pareto frontier, beating FlowDPS-on-RF in $5/6$ noisy cells while running $\sim$60$\times$ faster than DPS. We further evaluated thirteen publication-improvement candidates against a strict Bonferroni-corrected gate; *three* passed: `flowdps_rf_v2` (correctly EMA-loaded weights + power-law $\zeta$ ramp) adds $+0.77$ dB to FlowDPS-on-RF; `pixel_dps_spectral` (matched grid) adds $+0.46$ dB; and `flowdps_rf_spectral` (matched grid, $n{=}100$) adds $+0.54$ dB. The two spectral wins exhibit a paradoxical regime-reversal — the same algorithm fails under operator mismatch and wins under matched conditions — which we interpret as evidence that the underlying mechanism (downweight residuals in noise-dominated bands of the *assumed* forward operator) is correctly identified. The remaining ten candidates are documented as instructive near-misses or diagnosed failures, including a DPS-vs-$\Pi$-GDM L2-norm/squared convention mismatch, a guidance-saturation diagnosis from Heun at $2\times$ NFE, and a path-degeneracy diagnosis from gradient-free SMC. The findings give concrete guidance for choosing between these methods under known versus uncertain forward operators: prefer DPS when the operator is calibrated; consider flow-based posterior sampling when the operator may be mis-specified.

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